W-B-177a
May 12, 1964
SUPERSEDING
Fed. Spec. W-B-177
February 12, 1953

FEDERAL SPECIFICATION

BATTERY FILLER, GRAVITY

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers containers to dispense distilled water to the cells of storage batteries.

1.2 Classification.

1.2.1 *Types and styles.* Battery fillers furnished under this specification shall be of one type (jug) and the following styles, as specified (see 6.2):

Style A—With hose dispenser.
Style B—With valve control pouring spout.

2. APPLICABLE SPECIFICATIONS AND STANDARDS

2.1 The following specifications and standards, of the issues in effect on date of invitation for bids, form a part of this specification

Federal Specifications:

PPP-B-601—Boxes, Wood, Cleated, Plywood.

PPP-B-621—Boxes, Wood, Nailed and Lock-Corner.

PPP-B-636—Box, Fiberboard.

Federal Standards:

Fed. Std. No. 102—Preservation, Packaging and Packing Levels.

Fed. Std. No. 123—Marking for Domestic Shipments (Civilian Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C. 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D. C., Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, and Seattle, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specification:

MIL-P-116—Preservation, Methods of.

Military Standards:

MIL-STD-105—Sampling Procedures and Tables for Inspection by Attributes.

MIL-STD-123—Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

FSC 6140

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3. REQUIREMENTS

3.1 Standard product. The equipment shall be essentially the standard product of the manufacturer so that prompt and continuing service and delivery of spare parts may be assured. The component parts of the unit need not be the products of the same manufacturer.

3.2 Material. Material shall be as specified herein. Material not definitely specified shall be of a good commercial quality entirely suitable for the purpose. Material shall be free from all defects and imperfections that might affect the serviceability of the finished product.

3.2.1 The battery fillers shall be molded from a natural or synthetic rubber compound or a thermoplastic material capable of meeting the performance requirements of 3.6.

3.3 Construction.

3.3.1 *Style A fillers.* Style A fillers shall consist of a semi-closed shell, molded in one or two pieces, with an integral bail-type lifting handle at the top and an eye or buttoned-in holder for holding the nozzle end of the filling hose when not in use.

3.3.1.1 *Hose and fittings.*

3.3.1.1.1 *Hose.* The hose shall be attached not more than 1 inch from the bottom of the interior of the shell. The hose shall measure 18 inches ± 3 inches long with an inside diameter of 1/4 inch ± 1/16 inch. The dispensing end of the hose may have a small nonmetallic nozzle to restrict and direct the liquid flow. Such nozzle must not be affected by sulfuric acid or potassium hydroxide found in battery cells.

3.3.1.1.2 *Fittings.* Fittings in contact with battery water shall be of such materials as will not contaminate the water to be used for filling batteries containing sulfuric acid or potassium hydroxide.

3.3.2 *Style B fillers.* Style B fillers shall consist of a completely enclosed shell molded in one piece with an integral pitcher-type lifting handle and an integrally molded spout so positioned at the top as to provide ease of handling during the dispensing operation. The spout shall accommodate a removable control valve assembly which automatically limits the addition of water to the battery cells to prevent overfilling.

3.4 Capacity.

3.4.1 *Style A fillers.* Style A fillers shall have a minimum capacity of four quarts.

3.4.2 *Style B fillers.* Style B fillers shall have a minimum capacity of two quarts.

3.5 Weight. The empty battery fillers shall not exceed four pounds in weight.

3.6 Performance.

3.6.1 *Gravity drop.* The filler shall be subjected to the drop test of 4.4.1 without suffering any apparent injury to any part of the filler.

3.6.2 *Freezing.* The filler shall be subjected to the test of 4.4.2 without suffering any apparent injury to any part of the filler.

3.6.3 *Handle design.* The handle shall be so designed that it shall not suffer any apparent injury when the filler is subjected to test of 4.4.3.

3.6.4 *Temperature condition.* The filler shall be capable of satisfactory performance in any ambient temperature from 125° F. with full impact of solar radiation of 360 B.t.u/Ft. Sq./Hr. for at least 4 hours to 32° F. for at least 3 days without benefit of solar radiation, and shall not be damaged by storage (at any ambient temperature from 160° F. for periods of 4 hours daily to minus 80° F. for periods of at least 3 days duration.)

3.7 Identification marking. Each item of equipment shall bear the manufacturer's name or trademark on a nameplate securely

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affixed in a conspicuous place. In lieu of the nameplate, the manufacturer's name or trademark may be molded integral with, stamped, or otherwise permanently marked upon the item of equipment.

3.8 **Finish.** The equipment shall be finished in accordance with the manufacturer's standard practice for this type of equipment.

3.9 **Workmanship.** Workmanship shall be of the highest grade throughout and in accordance with good commercial practice for this type of equipment.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 **Responsibility for inspection.** Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 **Sampling.**

4.2.1 *Visual and mechanical inspection.* Sampling for visual and mechanical inspection shall be in accordance with MIL-STD-105, inspection level II, and acceptable quality levels (AQL) of 1.0 percent for major defects and 4.0 percent for minor defects.

4.2.2 *Test.* Sampling for test shall be in accordance with MIL-STD-105, inspection level S-4, AQL 6.5 percent.

4.3 **Inspection.** Battery fillers sampled in accordance with 4.2.1 shall be visually inspected to determine compliance with the requirements of this specification as follows:

Major defects:

Container material (3.2).
Container construction (3.3.1 and 3.3.2).
Hose nozzle material (3.3.1.1.1).
Fittings material (3.3.1.1.2).
Container weight (3.5).

Minor defects:

Hose dimensions (3.3.1.1.1).
Containers capacity (3.4)
Markings (3.7).
Finish (3.8).
Workmanship (3.9).

4.4 **Tests.** Battery fillers sampled in accordance with 4.2.2 shall be subjected to the following tests to determine compliance with the requirements of 3.6.

4.4.1 *Drop test.* The filler (and hose when applicable) shall be completely filled with small particles of sand and be subjected to a free fall from a height of 5 feet for three falls with the temperature of entire unit at 25°F.

4.4.2 *Freeze test.* The filler (and hose when applicable) shall be completely filled with water and be subjected to a temperature of 25° ± 2°F. for 4 hours and then warmed slowly to 40° ± 2°F. This shall be repeated three times.

4.4.3 *Handle strength test.* The filler (and hose when applicable) shall be completely filled with water and be subjected to a free fall of 3 feet from the end of a rope. This shall be repeated three times.

5. PREPARATION FOR DELIVERY

(For civil agency procurement the definitions and applications of the various levels of packaging and packing shall be as specified in Fed. Std. No. 102.)

5.1 **Packaging.** Packaging shall be level A, B, or C, as specified.

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5.1.1 *Level A.* The battery fillers shall be individually packaged in accordance with method IC-5 of MIL-P-116.

5.1.2 *Level B.* The battery fillers shall be individually packaged in snug fitting boxes conforming to PPP-B-636, type I, class 1. Suitable cushioning shall be provided to prevent movement within the boxes and each box shall be closed and sealed in accordance with the appendix to the box specification.

5.1.3 *Level C.* Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment from the supply source to the first receiving activity for immediate use.

5.2 **Packing.** Packing shall be level A, B, or C, as specified.

5.2.1 *Level A.* Packaged battery fillers shall be packed together in quantities not to exceed 200 pounds gross weight in snug fitting nailed wood boxes conforming to PPP-B-621, class 2, style 2, 2-1/2, or 3, or wood cleated plywood boxes conforming to PPP-B-601, style A or B, overseas type. Suitable cushioning shall be provided to prevent movement within the boxes. The boxes shall be closed and strapped in compliance with the appendix to the applicable box specification.

5.2.2 *Level B.* Packing shall be as outlined in 5.2.1, except domestic type boxes may be used.

5.2.3 *Level C.* The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.3 **Marking.**

5.3.1 *Civil agencies.* In addition to any special marking required in the contract or purchase order, shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.3.2 *Military agencies.* In addition to any special marking required in the contract or purchase order, shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 **Intended use.** The battery filler is intended for use in servicing automotive and truck electric storage batteries. Extreme caution should be exercised to prevent contamination of storage batteries through the use of the battery filler. In cases where a filler is used to service both acid and alkali type batteries all filler parts that come in contact with battery electrolyte should be washed thoroughly after servicing a particular battery type.

6.2 **Ordering data.** Purchasers should exercise any desired options offered herein, and procurement documents should specify the following:

(*a*) Title, symbol, and date of this specification.
(*b*) Style of filler (1.2.1).
(*c*) Inspection responsibility, if other than specified (4.1).
(*d*) Level of packaging and packing required (5.1 and 5.2).
(*e*) Special marking, if required (5.3).

6.3 This specification is believed to adequately describe the item under consideration. Normally no samples will be necessary prior to award to determine specification compliance. When samples of bids are believed necessary, both the request for the sample and the purpose thereby subserved should be specifically stated in the invitation for bids.

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6.4 Transportation description. Transportation description and minimum weights applicable to this commodity are:

Rail:

Battery filler, rubber.
Carload minimum weight 24,000 pounds, subject to Rule 34, Uniform Freight Classification.

Motor:

Truckload minimum weight 24,000 pounds, subject to Rule 115, National Motor Freight Classification.

MILITARY CUSTODIANS:

Army—None

Navy—None

Air Force—88

Review Interest:

EL, GL, MO, DOCKS, SHIPS, 88

User Interest:

MC

Copies of this specification may be purchased for cents each.

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